

STATISTICAL MODELLING OF THE SHRINKAGE BEHAVIOUR OF SOUTH AFRICAN CONCRETES

P. C. Gaylard¹

School of Statistics and Actuarial Science, University of the Witwatersrand, Private Bag 3,
WITS 2050, South Africa

e-mail: *petra.g@mweb.co.za*

and

L. P. Fatti

School of Statistics and Actuarial Science, University of the Witwatersrand, Private Bag 3,
WITS 2050, South Africa

and

Y. Ballim

School of Civil and Environmental Engineering, University of the Witwatersrand, Private Bag 3,
WITS 2050, South Africa

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Summary: A model for the time-dependent shrinkage profiles of South African concretes has been developed from historical laboratory data by hierarchical non-linear modelling. A database of all the available results of laboratory studies on concrete shrinkage carried out in South Africa was used to identify the most appropriate growth curve model for individual shrinkage profiles. The parameters of this growth curve model, fitted to each shrinkage profile individually, were then modelled in terms of suitable covariates, viz. the composition of the concrete, its other engineering properties as well as shrinkage test conditions. Use of this model to predict concrete shrinkage in South Africa is discussed.

1. Introduction

Concrete is one of the oldest and most widely used construction materials in the world. Concrete shrinks over time as it dries under normal atmospheric conditions, mainly from loss of moisture to the environment, but also from chemical reactions within the concrete, as well as reaction with carbon dioxide in the atmosphere (Kovler and Zhutovsky, 2006). Tensile stresses and, ultimately, cracks develop within the concrete if it is prevented from shrinking freely. Shrinkage is thus an important property of concrete as it influences the durability, aesthetics and long-term serviceability

¹Corresponding author

of the concrete as well as its load-bearing capacity (Owens, 2009). Hence, the accurate prediction of both the long-term shrinkage as well as the rate of shrinkage of concrete, as a function of its composition and environmental factors (such as temperature and humidity), is very important in the design stage of any concrete structure (American Concrete Institute, 2008).

Concrete is composed of hydraulic cement, supplementary cementitious materials (such as fly ash, blast furnace slag, condensed silica fume and limestone), coarse aggregate (stone – such as granite or quartzite), fine aggregate (sand – such as crushed stone or natural sand), water, and, in some cases, chemical admixtures. When these materials are mixed, the water reacts with the cement, forming hydration products that bond the other components together. The mixture eventually hardens to form a stone-like material.

There already exist a number of theoretical and empirical models to predict shrinkage (American Concrete Institute, 1982; Bažant and Baweja, 1996; Bažant, 2000; Fédération Internationale du Béton, 1999; Gardner, 2004; South African National Standard, 2000), but most do not take into account the effect of concrete raw materials, such as different supplementary cementitious materials and aggregate types. Furthermore, these models were all developed using data derived from non-South African concretes and thus do not take into account the effects of cements, supplementary cementitious materials and aggregates produced and used in South Africa, which may differ substantially from those used elsewhere. Studies on the effects of South African aggregates on concrete shrinkage have been carried out in the past (Marques, 1992; Alexander, 1993; Ballim, 1999; Ballim, 2000; Badenhurst, 2003) but these have not resulted in sufficiently reliable prediction models.

Shrinkage is defined as the decrease in volume of hardened concrete with time, when it is exposed to air of humidity less than 100% and is not subjected to loading or restraint (Owens, 2009). Shrinkage is defined as zero at the beginning of the drying time and is expected to reach an asymptote at large values of time (usually of the order of several years up to 30 years). Shrinkage measurements are taken at various time intervals by measuring the decrease in the distance between targets mounted on concrete specimens. Shrinkage at a given time t is then calculated as follows:

$$\text{Shrinkage at time } t = \frac{\text{change in length of sample at time } t}{\text{length of sample at start of test}} = -\frac{l(t) - l(t_0)}{l(t_0)}. \quad (1)$$

Since the length of the sample decreases with time, the negative sign is introduced to represent shrinkage as a positive quantity. Shrinkage is typically expressed as microstrain ($\times 10^{-6}$ m/m). A typical shrinkage profile for a concrete sample is shown in Figure 1.

2. Description of the data set and data preparation

The data for this study were provided by Ballim (School of Civil and Environmental Engineering, University of the Witwatersrand) and Alexander (Department of Civil Engineering, University of Cape Town) from a number of laboratory studies carried out between 1979 and 2009 in these institutions. The data set consisted of 245 experiments from 23 separate studies. An experiment is defined as an individual set of shrinkage measurements (as a function of time) together with information on concrete raw materials and composition as well as shrinkage testing conditions.

Each set of shrinkage measurements was taken by measuring the distance between two fixed points on each of two sides of three standard concrete prism samples which were being dried under

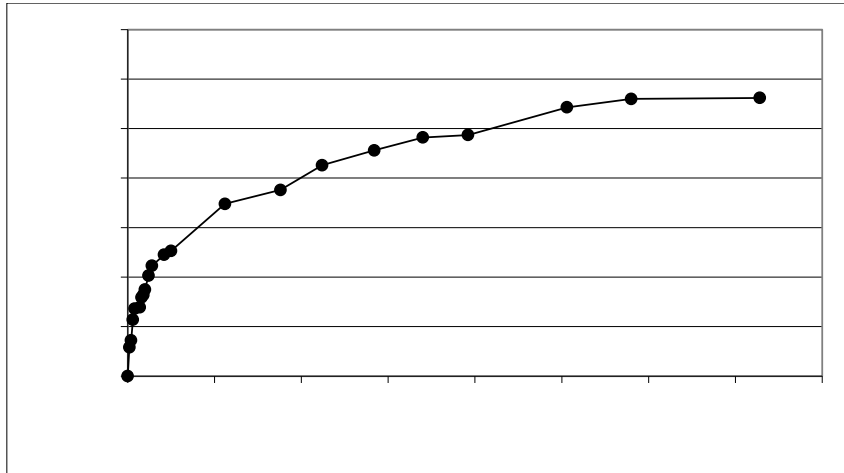


Figure 1: A typical shrinkage profile for a concrete sample.

controlled conditions of temperature and humidity. The measurement recorded at each time interval was the average of the six measurements taken on these three samples.

Use of historical data, gathered over a period of 30 years, for statistical modelling has a number of issues associated with it. Firstly, data gathered over such a long period of time are likely to be inconsistent, at least to some degree. The concrete mixes were made by different people, in several different laboratories, using different equipment and different batches of raw materials whose properties may not be consistent over time. The shrinkage measurements were carried out in two laboratories (with different specimen shapes and under slightly different conditions), by different people and using different equipment. In a planned experimental design, on the other hand, such variation would be controlled by means of blocking and factors not in the design would be held constant. However, for the data used in this study, the concrete preparation and shrinkage testing procedures are relatively straightforward and were carried out according to well-established procedures in reputable and well-controlled laboratories. The results could thus be expected to be fairly reproducible across the laboratories, equipment types and range of researchers preparing specimens.

Secondly, the individual studies making up the data set were not necessarily carried out with the development of a model for shrinkage as the main aim. Thus, not all important factors affecting shrinkage were varied over sufficiently wide ranges. In fact, some factors were kept constant because they were known to have a significant effect on shrinkage, e.g. the level of humidity during the shrinkage test. However, a model can still be developed with this limitation in mind, and can then be enhanced by further, designed, experiments to include such factors. Further, not all data required for this study were recorded in some of the studies making up the data set. This missing data were regarded as Missing Completely At Random (MCAR). Except for the properties of the aggregates (see later) no replacement of missing data was carried out.

Thirdly, collinearity may exist between variables whose levels were not changed independently (as would be the case in an orthogonal experimental design). However, given the different studies

making up the data set there was sufficient variation in the concrete mixes represented so that this was not problematic.

Fourthly, lurking variables and serially correlated errors which, in a designed experiment, would have been accounted for by randomisation and blocking, could be problematic in the analysis of historical data. However, in this case, the data set was made up of a number of unrelated studies (subsets of data) which provided some insurance against possible bias in the data resulting from the lack of blocking and randomisation.

Finally, not all levels of all variables were equally represented in the data set. For example, sixty experiments were carried out using dolerite as the stone type, which means that results for this stone type can be interpreted with greater confidence compared to those for other stone types represented by fewer experiments.

Despite these concerns, the enormous variety of cement and aggregate types as well as concrete compositions covered by the data set presented an unprecedented opportunity to discover the important concrete composition factors which influence shrinkage and thus to provide a basis for further, designed, experiments.

The data as provided contained 42 time-independent covariates, describing cement and aggregate type, concrete mix composition, concrete curing conditions, shrinkage specimen dimensions, shrinkage test conditions and the results of other tests carried out on the concrete specimens. There were no time-dependent covariates. Further details are given in Gaylard (2011).

Cement type and concrete composition definitions were standardized across the data set. Data on the physical properties of the aggregates were collated, generating a further fifteen time-independent covariates, with the hope of describing the aggregates by means of one or more continuous properties, as opposed to simply indicating their type by using indicator variables. Ratios of some key variables were defined as further covariates.

The shrinkage data for one experiment comprised repeated measures on the same sample. The times at which the measurements are taken were not separated by constant intervals, nor were the times necessarily the same from experiment to experiment. The shape of the shrinkage profile resembled a growth curve. Some shrinkage experiments were conducted for much longer periods (up to approx. 900 days) than others (some as low as approx. 28 days). Furthermore, some shrinkage experiments ended before the shrinkage started to approach an asymptote, as illustrated in Figure 1.

3. Overall approach to developing a model for shrinkage from the data set

The following characteristics of the data and requirements of the model were established:

- The data consist of repeated response (shrinkage) measurements over time, made on a number of different experiments.
- The shrinkage y depends nonlinearly on a set of unknown parameters β for each experiment. At the very least, the relationship between shrinkage and time is nonlinear.
- The shrinkage profiles are similarly shaped across experiments, but may have different values of the parameter vector β for different experiments.

- The inter-experiment variation between the values of the parameter vector $\boldsymbol{\beta}$ may be considered to be a combination of experiment-specific characteristics (i.e. related to the covariates) and random variability.
- As far as possible, the model's form and parameters should have a practical interpretation.

The approach thus used in modelling the data was that of hierarchical nonlinear modelling (Davidian and Giltinan, 1995). This approach is suitable in situations where the data consist of repeated observations of a continuous response (shrinkage) on each of several subjects (the individual experiments) over a continuous variable (time); where there is variability in the relationship between the response and time across subjects; and where parameters (covariates) exist which vary across experiments and characterise the behaviour of individual subjects, dictating the variation in the patterns of time-response. Hierarchical nonlinear modeling may be used to model the typical behaviour of the phenomena represented by the covariates: the extent to which the covariates vary across experiments and whether some of the variation is systematically associated with individual attributes. Using the model, individual-level prediction may also be done (Davidian and Giltinan, 2003). This approach has been used in a variety of applications (Davidian and Giltinan, 2003; Yeap, Catalano, Ryan and Davidian, 2003).

The general outline given by Davidian and Giltinan (1995) is followed. Let y_{ij} denote the shrinkage at time t_{ij} , $j = 1, \dots, n_i$, for the i^{th} experiment, $i = 1, \dots, m$. A nonlinear function $f(\mathbf{t}, \boldsymbol{\beta})$ models the relationship between the response at time $\mathbf{t}$, where $\boldsymbol{\beta}$ is a $(p \times 1)$ vector of unknown parameters. The function f has the same basic functional form for all the experiments and depends on the regression parameters $\boldsymbol{\beta}_i$ specific to the i^{th} experiment. Thus

$$E(y_{ij}) = f(t_{ij}, \boldsymbol{\beta}_i). \quad (2)$$

The *first stage of the model* describes the systematic and random features of the data at the *intra-experiment* level. For the i^{th} experiment, the j^{th} response can be written as

$$y_{ij} = f(t_{ij}, \boldsymbol{\beta}_i) + e_{ij} \quad (3)$$

where e_{ij} is a random error term reflecting the residual variability in the response, over and above that explained by the model, in the i^{th} experiment, with $E(e_{ij}) = 0$. The data for the i^{th} experiment can thus be summarised as follows:

$$\begin{aligned} \begin{bmatrix} y_{i1} \\ \vdots \\ y_{in_i} \end{bmatrix} &= \begin{bmatrix} f(t_{i1}, \boldsymbol{\beta}_i) \\ \vdots \\ f(t_{in_i}, \boldsymbol{\beta}_i) \end{bmatrix} + \begin{bmatrix} e_{i1} \\ \vdots \\ e_{in_i} \end{bmatrix} \\ \mathbf{y}_i &= \mathbf{f}_i(\boldsymbol{\beta}_i) + \mathbf{e}_i \end{aligned} \quad (4)$$

where $E(\mathbf{e}_i) = 0$. The assumption that the e_{ij} are independent with constant variance was found to hold reasonably in practice.

In the *second stage of the model*, *inter-experiment variation* is characterised by a model for variation in each of the p regression parameters in $\boldsymbol{\beta}$. This is done by deriving a multiple regression model for each of the p regression parameters in the first stage model:

$$\boldsymbol{\beta}^{(s)} = \mathbf{A}^{(s)} \mathbf{b}^{(s)} + \boldsymbol{\delta}^{(s)} \quad (5)$$

where

- $s = 1 \dots p$.
- $\boldsymbol{\beta}^{(s)}$ is an $(m \times 1)$ vector containing the values of the $\hat{\beta}_{si}$ estimated for each of the m experiments in the first stage (or a transformation of the $\hat{\beta}_{si}$).
- $\mathbf{A}^{(s)}$ is an $(m \times (k+1))$ matrix containing the values of the k covariates for the m experiments, plus a column (to refer to the intercept) containing m values equal to 1.
- $\mathbf{b}^{(s)}$ is a $((k+1) \times 1)$ vector containing all the model parameters to be estimated on the basis of the data: the intercept and k slope coefficients relative to each independent variable.
- $\boldsymbol{\delta}^{(s)}$ is an $(m \times 1)$ vector containing the error terms.

For a given set of experimental conditions (covariate), each regression parameter may then be estimated on the basis of its multiple regression model (eqn. 5). Once each regression parameter has been estimated, these estimates may be substituted into the nonlinear regression model to obtain a prediction of the shrinkage profile for the given set of experimental conditions.

The main steps in developing the model were thus the following:

- **First stage**
 - ▶ Modelling the individual shrinkage profiles by means of growth curves and nonlinear regression,
 - ▶ choosing the best growth curve model for the data,
- **Second stage**
 - ▶ exploratory data analysis and weighting of data in preparation for modelling of the effects of the covariates,
 - ▶ modelling the parameters of the chosen growth curve in terms of the covariates, and
 - ▶ assessing the goodness-of-fit of the models to the data and choosing the best model for the data.

4. Modelling the individual shrinkage profiles by means of non-linear regression

Of the many growth curve models available (Ratkowsky, 1990) those considered for fitting to individual shrinkage profiles were selected according to the following criteria: for a growth curve model, $y = f(t)$,

- the shape of the growth curve should resemble that of the shrinkage profile shown in Figure 1,

- $f(t) = 0$ at $t = 0$,
- $\lim_{t \rightarrow \infty} f(t) = \text{ultimate shrinkage} = \text{constant}$ (the constant is a parameter to be modelled in the second stage of the hierarchical modelling process),
- the parameters should have a practical meaning if at all possible, i.e. they should relate to the ultimate shrinkage, rate of shrinkage, etc., and
- the number of parameters should encourage model parsimony (Ratkowsky, 1990).

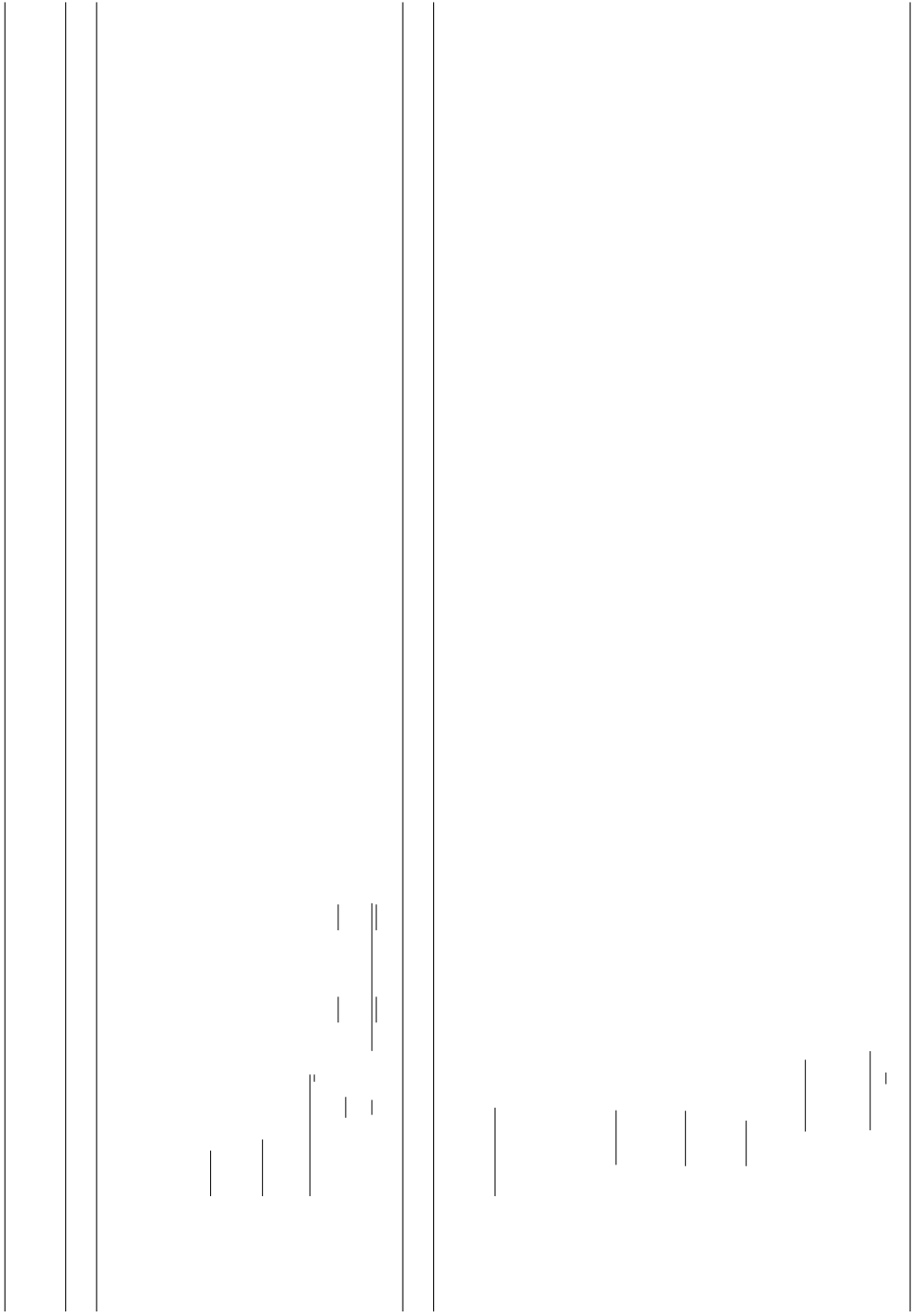
In total, fifteen suitable candidate growth curves were identified, according to the above criteria, from various literature sources. The models and their sources are listed in Table 1.

A subset of ten “key” shrinkage profiles was selected from the 245 profiles in the data set, to which all fifteen growth curve models were fitted. These “key” profiles, identified in Figure 2 by their profile numbers (#), were selected according to criteria chosen to identify the best quality shrinkage data spanning a range of ultimate shrinkage and shrinkage rate values. All fifteen growth curve models were fitted to each of the ten selected shrinkage profiles, using OLS estimation. Figures 2 and 3 illustrate the fit of one model (Model 10) to all ten profiles, and the fit of all fifteen growth curve models to one shrinkage profile, respectively. Models 3 and 6 proved to be inappropriate to describe the shrinkage profiles, as illustrated in Figure 3 (a), and were excluded from further consideration.

None of the thirteen remaining growth curve models was found to perform consistently poorly on the standard graphical diagnostics on all ten shrinkage profiles and thus no further models were eliminated on this basis. In order to determine which of the remaining thirteen growth curve models gave the best fit to the shrinkage profiles, the following goodness-of-fit measures were used: Pseudo- R^2 (UCLA Academic Technology Services, 2008), the Concordance Correlation Coefficient (CCC) between the observed and predicted values (Vonesh, Chinchilli and Pu, 1996) (adapted for a single shrinkage profile), AIC and BIC (Piegorsch and Bailer, 2005), and the Root Mean Square (RMS) (Motulsky and Ransnas, 1987).

For each model, average values (across the ten shrinkage profiles) for each of the five goodness-of-fit measures were calculated. These results are presented in Table 2. The penalization on moving from 2-parameter models (models 1-7) to 3-parameter models (models 8-15) can clearly be seen in the AIC values whereas the other four goodness-of-fit measures showed that the three-parameter models gave better fits than the two-parameter models. Those parameters which gave values on an absolute scale (0 to 1), viz. the pseudo- R^2 and the CCC, indicated that all the models fitted fairly well (the lowest values were 0.966 and 0.984 for these two measures respectively).

Models 10, 12, 13 and 15 appeared to give the best fit, as determined by the five goodness-of-fit measures, but a ‘best model’ could not be identified easily. Two Multiple Criteria Decision Analysis (MCDA) approaches were selected for the identification of the most suitable growth model for the shrinkage-time profiles, viz. the Value Function and Outranking methods.



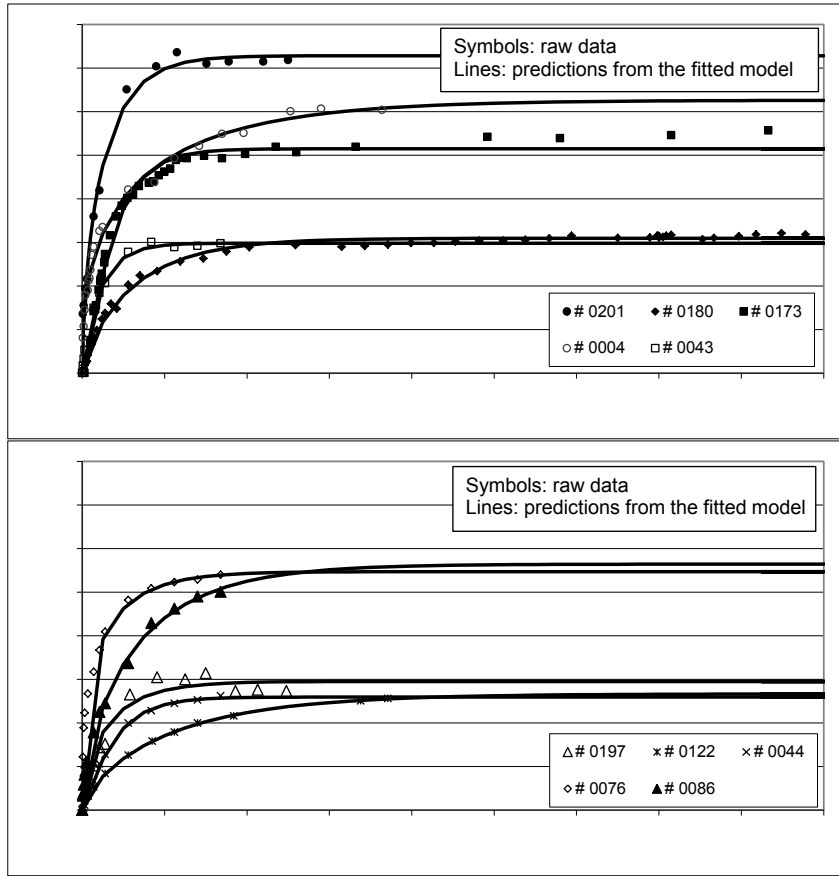


Figure 2: Illustration of the fit of one growth curve model (Model 10) to the ten “key” shrinkage profiles.

The implementation of the Value Function method was based on the guidelines given by Belton and Stewart (2002). Since only the pseudo- R^2 and CCC criteria had defined ranges (0 to 1) but their actual values were clustered near the upper end of these ranges, it was decided to use local scaling for all five criteria. Equal weights were assigned to all criteria. The overall value functions for the thirteen growth curve models are compared in Figure 4. Models 10, 12, 13 and 15 thus gave the best fits to the “key” shrinkage profiles, after which there was a substantial decrease in value function scores for the rest of the models. However, a decision for one “best” model could not be made on the basis of the value measurement results since the top two models had identical value function scores.

The particular method by which the Outranking relationship was defined and implemented in this study was the ELECTRE I method (Roy, 1996), the earliest and simplest of outranking methods, as presented by Belton and Stewart (2002). For the concordance index, the five goodness-of-fit criteria were weighted equally, as before. For the discordance index, twice the average standard error for each criterion (averaged over the thirteen models) was used as the veto threshold. The concordance

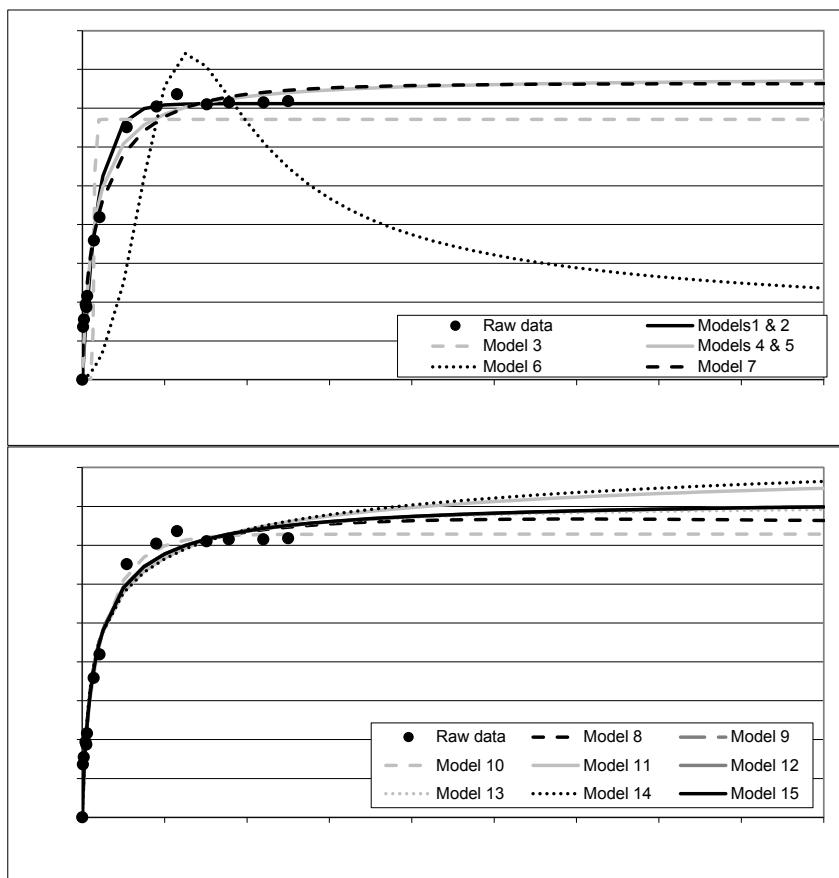


Figure 3: Illustration of the fit of the (a) two-parameter and (b) three-parameter growth curve models to one shrinkage profile.

threshold was set at $C^* = 0.7$ (after some experimentation to give the clearest separation between the models) and the discordance threshold was set at $D^* = 0.1$ (the discordance index can take on values of only 0 or 1). The resulting outranking relationship between the thirteen growth curve models is depicted in Figure 5.

The results showed that the same four models, viz. Models 10, 12, 13 and 15, were ranked as the best performing models. In contrast to the value function method, Models 10 and 13 were separated by the outranking method: Model 10 outranked Model 13. Models 15 and 12 followed in the same order as observed in the value function method. There was a large number of incomparable (as well as indifferent) pairwise comparisons lower down in the outranking relationship (see Figure 5), but this was not of concern since the aim was to identify only the best methods.

After fitting the four best growth curve models to *all* the shrinkage time profiles, Model 10 was chosen as the ‘best’ model on the basis of model convergence, confidence intervals for parameter estimates, and stability of long-term shrinkage prediction, particularly for those shrinkage profiles

Table 2: Goodness-of-fit scores for each growth curve model. For each goodness-of-fit statistic, the models with the first, second and third best scores are highlighted in dark, medium, and light grey, respectively.

Model	Goodness-of-fit score (95% confidence interval)				
	AIC	BIC	Pseudo-R ²	CCC	RMS
Model 1	10.0 (± 1.3)	128 (± 54)	0.966 (± 0.027)	0.984 (± 0.012)	27.0 (± 14.2)
Model 2	10.0 (± 1.3)	128 (± 54)	0.966 (± 0.027)	0.984 (± 0.012)	27.0 (± 14.2)
Model 4	9.7 (± 1.2)	121 (± 52)	0.978 (± 0.015)	0.989 (± 0.008)	21.9 (± 9.9)
Model 5	9.7 (± 1.2)	121 (± 52)	0.978 (± 0.015)	0.989 (± 0.008)	21.9 (± 9.9)
Model 7	10.1 (± 0.7)	128 (± 62)	0.974 (± 0.016)	0.986 (± 0.009)	23.9 (± 8.4)
Model 8	11.0 (± 1.2)	108 (± 49)	0.988 (± 0.014)	0.993 (± 0.007)	16.4 (± 9.1)
Model 9	11.0 (± 1.4)	111 (± 53)	0.986 (± 0.014)	0.994 (± 0.007)	17.2 (± 9.0)
Model 10	10.6 (± 1.3)	105 (± 56)	0.991 (± 0.010)	0.996 (± 0.005)	13.3 (± 6.5)
Model 11	11.0 (± 1.0)	104 (± 40)	0.988 (± 0.014)	0.994 (± 0.007)	15.9 (± 9.4)
Model 12	10.7 (± 1.3)	100 (± 44)	0.990 (± 0.013)	0.995 (± 0.007)	14.4 (± 8.5)
Model 13	10.7 (± 1.2)	99 (± 43)	0.990 (± 0.012)	0.995 (± 0.006)	13.9 (± 7.8)
Model 14	11.2 (± 1.0)	107 (± 40)	0.986 (± 0.015)	0.993 (± 0.008)	17.5 (± 9.9)
Model 15	10.6 (± 1.3)	99 (± 44)	0.990 (± 0.013)	0.995 (± 0.007)	14.2 (± 8.3)

with insufficient long-term data. As described in Table 1, Model 10 is given by

$$\text{Shrinkage}(t - t_0) = \alpha[1 - e^{-\beta(t-t_0)}]^\gamma, \quad \alpha, \beta, \gamma > 0, \quad (6)$$

where α represents the ultimate shrinkage, β the growth rate and γ the shape factor.

Finally, in order to complete the discussion on growth curve model selection, it was noted that the graphical diagnostics for the chosen model were acceptable. The residuals were not appreciably correlated with time nor with each other and thus the assumption of an uncorrelated intra-experiment error structure, made in Section 3, was justified.

5. Modelling the parameters of the growth curve model in terms of the covariates

The results of the fitting of the growth curve model to the individual shrinkage profiles showed that the estimates of the three parameters were generally highly correlated, particularly the parameters β and γ . As a result it was decided to model the three parameters together by means of multivariate multiple regression (Johnson and Wichern, 1988; Rencher, 1998) rather than modelling them individually by simple multiple regression. In order to derive a reliable second-stage model, it was necessary to exclude experiments for which data were insufficient to allow a realistic prediction of the ultimate shrinkage, i.e. for which either

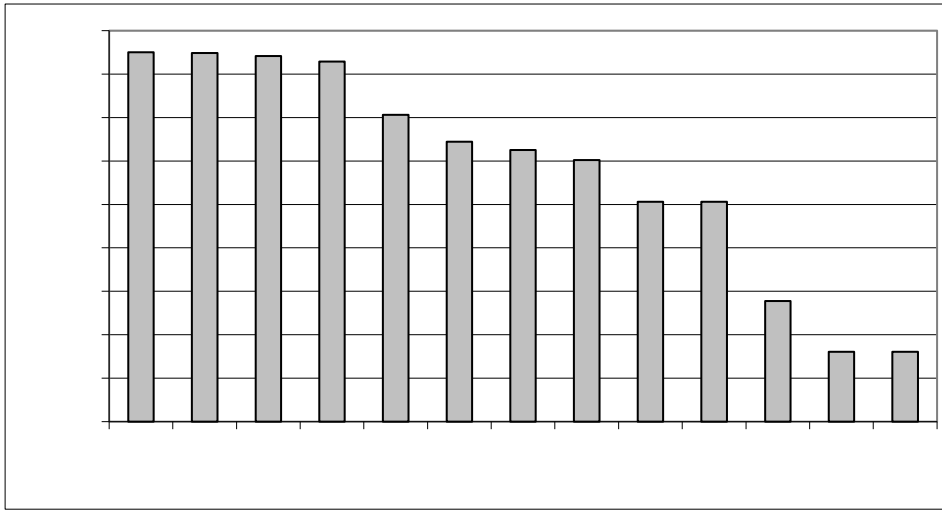


Figure 4: Comparison of the value functions for the growth curve models.

- the shrinkage measurements ceased before 60 days, which is the minimum drying time specified for the natural shrinkage test method (American Society for Testing and Materials, 1999) or
- the shrinkage predictions continued to increase substantially at very long drying times. Such profiles were identified by comparing the estimated value of α (the ultimate shrinkage) with the value of the last shrinkage measurement. If these differed by more than 15% (a limit set by inspection), the experiment was excluded from the regression analysis.

The total number of experiments retained was 174, representing 71% of the original data set of 245 experiments.

While the distribution of the dependent variable α was close to normal, the distribution of β was found to be significantly positively skewed, while that of γ was found to be both positively skewed and leptokurtic. The application of a natural logarithm transformation to both these variables reduced these problems to a large extent. It was thus decided that the dependent variables to be used in the multivariate multiple regression would be α , $\ln(\beta)$ and $\ln(\gamma)$.

Covariates with more than 30% missing data were excluded, which was judged to not seriously affect the quality of the data set.

Bivariate plots and the correlation matrix were used to identify highly correlated relationships between the covariates. The covariates were then grouped into sets for evaluation by multivariate multiple regression. The sets were constructed so that each set would adequately describe the composition and properties of the concrete, while avoiding the collinear relationships between the variables described above. Each set of covariates would then be subjected to variable selection as part of the multivariate multiple regression to find that subset of covariates within each set which best described the variability in the dependent variables. The models thus derived from each set of

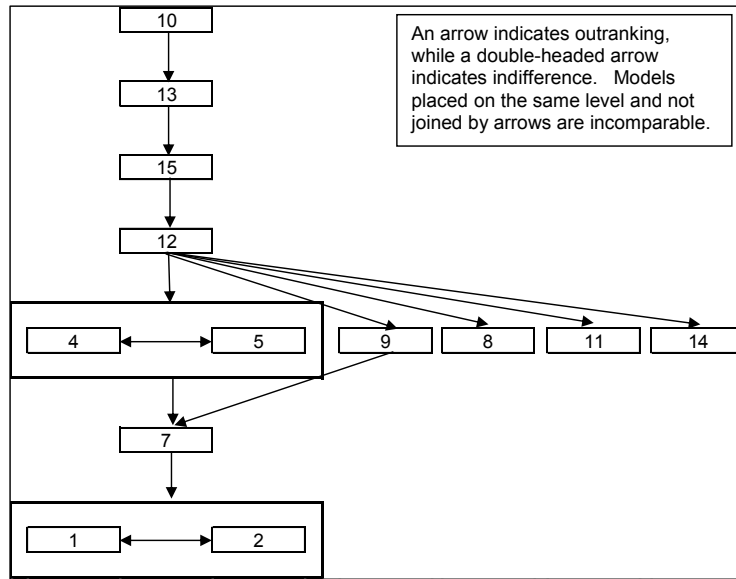


Figure 5: The outranking relationship for the thirteen growth curve models according to the ELECTRE I method.

covariates would then be compared to find the overall best model.

Forward and backward stepwise multivariate regression analyses were used for variable selection within each of the 768 groups of covariates used as starting points for the multivariate multiple regressions. This approach offered the greatest flexibility in dealing with the large number of covariates in many of the covariate sets. Both weighted and unweighted regressions were performed; weights were allocated in proportion to the number of data points contained in each shrinkage profile. From all these regressions, 11 promising models were identified based on the final average adjusted R^2 (for the three dependent variables). These models were assessed by goodness-of-fit measures and the best model was selected using MCDA. The following goodness-of-fit criteria were used: the multivariate final prediction error (MFPE), the Corrected Akaike's Information Criterion (CAIC), BIC, the Corrected Hannan and Quinn (CHQ) statistic (Kim and Timm, 2007), and the average adjusted R^2 (for the three dependent variables). The results are presented in Table 3.

The unweighted models had lower CAIC, BIC and CHQ scores than the weighted models, while the weighted models had higher average adjusted R^2 values than the unweighted models. There was no one model which had the best scores across all five goodness-of-fit measures, although Model 17* had the lowest scores for CAIC, BIC and CHQ. Therefore, selection of the best model was carried out by MCDA, based on the five goodness-of-fit measures. The Value Function and Outranking methods, described in Section 4, were used once again.

For the Value Function method, only the average adjusted R^2 criterion had a defined range (0 to 1), but since the actual values of this statistic were clustered in a narrow range (0.66-0.70), it was decided to use local scaling for all criteria. Equal weights were assigned to all criteria. The overall

Table 3: Goodness-of-fit scores for the eleven multivariate multiple regression models. For each goodness-of-fit statistic, the models with the first, second and third best scores are highlighted in dark, medium, and light grey, respectively.

Model	Number of cases	Number of parameters	Goodness-of-fit score				
			MFPE	CAIC	BIC	CHQ	Average adjusted R^2
Model 2*	166	21	8.74	2.789	585.87	3.269	0.660
Model 3*	166	21	9.17	2.837	593.89	3.317	0.664
Model 4*	166	20	8.96	2.781	580.81	3.230	0.667
Model 5*	166	20	9.12	2.800	583.82	3.248	0.661
Model 6*	166	20	9.03	2.789	582.09	3.238	0.662
Model 12*	166	19	9.51	2.809	581.38	3.228	0.658
Model 17*	165	20	8.43	2.725	568.15	3.176	0.666
Model 18*	165	22	8.10	2.753	580.04	3.265	0.664
Model 19*W	167	25	8.79	2.929	626.39	3.530	0.691
Model 20*W	167	25	8.34	2.877	617.76	3.478	0.697
Model 21*W	167	25	8.31	2.873	617.05	3.474	0.698

value functions are compared in Figure 6.

Model 17* obtained the highest value function score, followed by Model 18*, after which there was a considerable decrease in value function scores for the rest of the models. Application of the Outranking method gave the same ordering for the top three models. Model 17*, now known as the WITS model, was thus chosen as the final model.

6. Description of the WITS model

The cement, stone, and sand types covered by the WITS model are limited to the cement, stone, and sand types and levels included in the data used in the final multivariate multiple regression analysis and also their combinations. Further details may be found in our companion publication (Gaylard, Ballim and Fatti, 2013). The coefficients of the model terms for the three dependent variables are given in the Appendix.

The final step in the description of the model was to compute 95% prediction intervals for the shrinkage-time profiles, derived from the two-stage model. Both the Delta Method (Weisberg, 2005) and simulation (Chernick, 2008) gave very similar standard deviations for the predicted shrinkage. Assuming approximate normality (achieved earlier by the log transformations applied to the parameters β and γ), the required prediction intervals were calculated. To give an overview of the fit of the model to the raw data, the two shrinkage profiles with predicted values of α closest to the observed asymptote as well as the two shrinkage profiles with predicted values of α furthest from the observed asymptote, from the cases in the multivariate multiple regression data set, are shown in Figure 7.

The model has been interpreted in relation to what is known about the shrinkage behaviour of

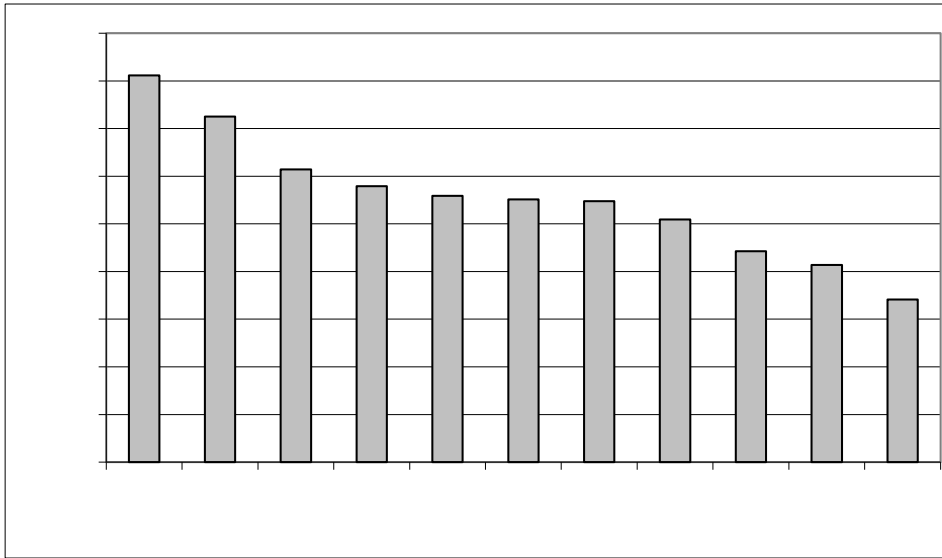


Figure 6: Comparison of the value function scores for the eleven multivariate multiple regression models.

concrete, and was found to conform well with practical experience in most respects.

A final aim of the study was to compare the model developed in this study to several models for shrinkage already in use in the concrete industry, with respect to the data in the South African database. The results of this comparison are reported in a separate publication (Gaylard et al., 2013). It was found that the WITS model performed the best across a variety of graphical and numerical goodness-of-fit criteria.

7. Conclusion

The importance of the study is two-fold: Firstly, the concept of hierarchical nonlinear modelling, has been applied for the first time to the modelling of concrete shrinkage. This approach could prove useful to other researchers seeking to model concrete shrinkage and related time-dependent properties such as creep. Secondly, the model derived in this study is the first comprehensive model to bring together laboratory data on the shrinkage of concrete generated in South Africa over a span of 30 years, identifying which covariates are the most important contributors to both the magnitude and rate of concrete shrinkage. Recognising the limitations of the study, the model provides a starting point for further, statistically designed, experiments to explore the effect of key variables more fully.

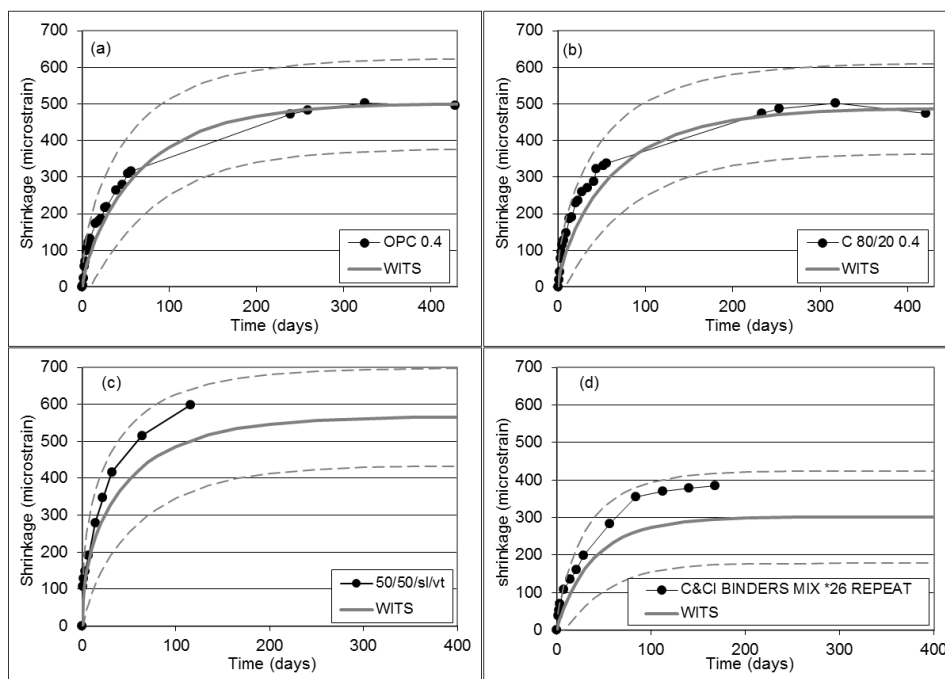


Figure 7: Predictions and 95% prediction intervals for the two shrinkage profiles with predicted values of α ((a) and (b)) closest to the observed asymptote and ((c) and (d)) furthest from the observed asymptote.

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Appendix: Summary of the WITS model for concrete shrinkage

Mean shrinkage strain in the cross-section: $\varepsilon_{sh}(t-t_0) = \alpha(1 - e^{-\beta(t-t_0)})^\gamma$	
$\alpha = -2245.19$ + Cement type factor (choose one cement type): -3.85 CEM V A 0 CEM I, CEM II A-D, CEM II A-L, CEM II A-M(L), CEM II A-S, CEM II A-V, CEM II B-M(V/L), CEM II B-S, CEM II B-V, CEM III A GGCS and CEM III A GGFS 8.63 CEM III A GGFS + Stone type factor (choose one stone type): -43.02 Granite 0 Andesite, Dolerite, Dolomite, Greywacke, Pretoria Quartzite, Shale, Wits Quartzite 211.75 Tillite 302.21 Quartzite + Sand type factor (choose one sand type unless given proportions indicate otherwise): 0 Cape Flats, Dolomite, Ecce Grit, Pretoria Quartzite, Quartzite (0 to 80%), River (0 to 25%), Tillite (0 to 80%) 99.54 Andesite 134.27 Dolerite 139.92 Klipheuwel Pit 170.57 Natural 201.75 Granite 260.15 Wits Quartzite 321.77 Shale 431.08 River Vaal (0 to 20%) + 55.71 * ln(Cement Content in kg/m³) + 2.54 * Water Content in kg/m³ - 0.05 * Stone Content in kg/m³ + 25.35 * Aggregate / Binder mass ratio + 173.17 * ln(2*Volume To Surface Area Ratio in mm) + 44.34 * Temperature in °C	$\ln(\beta) = 9.76$ + Cement type factor (choose one cement type): -0.29 CEM III A GGFS 0 CEM I, CEM II A-D, CEM II A-L, CEM II A-M(L), CEM II A-S, CEM II A-V, CEM II B-M(V/L), CEM II B-S, CEM II B-V, CEM III A GGCS and CEM III A GGFS 0.79 CEM V A + Stone type factor (choose one stone type): -0.32 Tillite 0 Andesite, Dolerite, Dolomite, Greywacke, Pretoria Quartzite, Shale, Wits Quartzite, Quartzite 0.33 Granite + Sand type factor (choose one sand type unless given proportions indicate otherwise): -0.64 Wits Quartzite -0.49 Shale -0.42 Natural -0.36 Granite -0.35 River Vaal (0 to 20%) 0 Cape Flats, Dolomite, Ecce Grit, Pretoria Quartzite, Quartzite (0 to 80%), River (0 to 25%), Tillite (0 to 80%) 0.04 Andesite 0.43 Klipheuwel Pit 1.22 Dolerite + 0.01 * ln(Cement Content in kg/m³) - 0.05 * Aggregate / Binder mass ratio - 1.76 * ln(2*Volume To Surface Area Ratio in mm) - 0.26 * Temperature in °C
Ranges covered by the model: Cement Content: 112-536 kg/m³ Water Content: 160-225 kg/m³ Aggregate / Binder Ratio: 3.18-8.74 Stone Content: 900-1400 kg/m³ 2*Volume To Surface Area Ratio: 16.5-75.0 Temperature: 21-25°C Humidity: 43-72% For combinations of cement, stone and sand covered by the model, see Gaylard (2013). Abbreviations: GGBS Ground Granulated Blast-furnace Slag GGCS Ground Granulated Corex Slag GGFS Ground Granulated Ferro-manganese Slag	$\ln(\gamma) = 3.04$ + Cement type factor (choose one cement type): -0.01 CEM III A GGFS 0 CEM I, CEM II A-D, CEM II A-L, CEM II A-M(L), CEM II A-S, CEM II A-V, CEM II B-M(V/L), CEM II B-S, CEM II B-V, CEM III A GGCS and CEM III A GGFS 0.19 CEM V A + Stone type factor (choose one stone type): 0 Andesite, Dolerite, Dolomite, Greywacke, Pretoria Quartzite, Shale, Wits Quartzite 0.01 Quartzite 0.05 Tillite 0.34 Granite + Sand type factor (choose one sand type unless given proportions indicate otherwise): -2.58 River Vaal (0 to 20%) -0.44 Wits Quartzite -0.40 Shale -0.36 Granite -0.12 Natural -0.03 Andesite 0 Cape Flats, Dolomite, Ecce Grit, Pretoria Quartzite, Quartzite (0 to 80%), River (0 to 25%), Tillite (0 to 80%) 0.02 Klipheuwel Pit 0.50 Dolerite + 0.16 * ln(Cement Content in kg/m³) + 0.08 * Aggregate / Binder mass ratio - 0.62 * ln(2*Volume To Surface Area Ratio in mm)

